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Palaeomagnetism and K-Ar dating of Pleistocene volcanic rocks along the Altyn Tagh fault, northern border of Tibet

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SUMMARY

Two basaltic andesite lava flows were sampled in the Puru area (36°15′N; 81°30′E) north of the Altyn Tagh fault. The sampling sites were distributed between 13 km and 18 km from the fault. The two lavas are dated by K-Ar methods to be 1.05 ± 0.11 and 1.44 ± 0.06 Ma. Their palaeomagnetic directions show reversed polarity of the Matuyama chron. Declination values for nine sampling sites on the upper lava flow more than 14 km from the fault are concordant, with little variance ($D = -166.3°$, $I = -50.8°$, $\alpha_{95} = 2.2°$), but declination at a site nearer the fault (13 km from it) ($D = -173.7°$) displays a counter-clockwise deflection of $7.4° \pm 5.2°$ from that of the other sites. This palaeomagnetic rotation is ascribed to tectonic rotation due to left-lateral slip on the Altyn Tagh fault during the last 1.0 myr. The area beyond 13 km from the fault, however, does not provide any evidence for the large rotation (~60°) expected from the known strike-slip movement of 30 mm yr^{-1}. This implies that little deformation penetrates deep into the Tarim craton from the fault. The lack of deformation of features along the Altyn Tagh fault is explained by the non-rigid behaviour of the narrow zone of the continental lithosphere near the fault and/or by low stress acting on the fault.

Key words: Altyn Tagh fault, K-Ar dating, palaeomagnetism, rotation, strike-slip fault, tectonics.

INTRODUCTION

Palaeomagnetic evidence for large-scale tectonic deformation in the Asian continent has been accumulated during the last decade (Achache, Courtillot & Zhou 1984; Chen *et al.* 1993; Huang & Opdyke 1993). Continents on plates have consequently been recognized to be not perfectly rigid. One of the features of non-rigid deformation is found in large-scale continental strike-slip zones. Displacement along strike-slip faults brings about crustal rotation in tectonic zones near the faults (England, Houseman & Sonder 1985; Luyendyk *et al.* 1985; Sonder, England & Houseman 1986). We have studied deformation features near a strike-slip fault using the palaeomagnetic method to shed light on the nature of the continental lithosphere.

Many large strike-slip faults are distributed around the Tibetan Plateau in the Asian continent, e.g. the Altyn Tagh, Karakorum, Kunlun and Red River Faults (Molnar & Tapponnier 1975; Tapponnier *et al.* 1982) (Fig. 1). We focused on the Altyn Tagh fault in the Tibetan Plateau to detect crustal rotation, because it is one of the world's greatest strike-slip faults (Molnar & Tapponnier 1975; Peltzer, Tapponnier & Armijo 1989). The Altyn Tagh fault follows the northern margin of the Tibetan highlands for nearly 2000 km. The rate of the Holocene left-lateral slip of the Altyn Tagh has been estimated to be 30 ± 20 mm per year from the Quaternary offsets of moraines and erosional features (Peltzer, Tapponnier & Armijo 1989).

We have collected samples from Pleistocene basaltic andesite lava flows near Puru village (36°08′N; 81°26′E) (Fig. 1). We have measured the palaeomagnetism and K-Ar ages of these lavas in an attempt to examine the rotation properties associated with deformation along the fault.

GEOLOGICAL SETTING AND SAMPLING

The Puru area lies on the southern margin of the Tarim Craton, which is bounded to the south by the Songban–Ganzi Terrane (Dewey *et al.* 1988). The Kunlun–Qinling Suture marks the boundary between the two terranes. It marks the southern limit of the composite Eurasian plate before the accretion of the Songban–Ganzi Terrane in the late Triassic. The Altyn Tagh fault trends approximately parallel and adjacent to this suture.

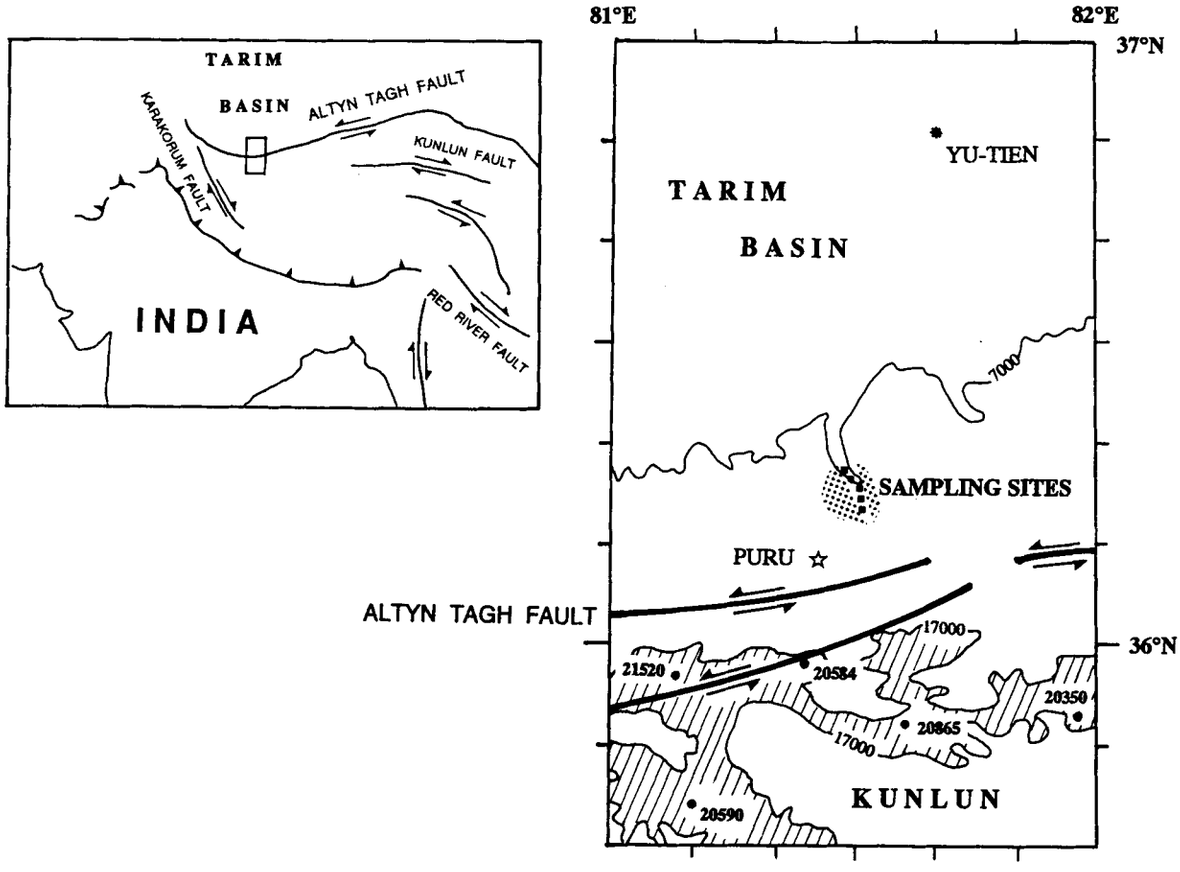


Figure 1. Location map showing some features of the geography and geology around Puru. Sampling sites are shown by a chain of squares. Heavy lines indicate the Altyn Tagh fault after Peltzer, Tapponnier & Armijo (1989). The dotted area shows outcrop of the Pleistocene basaltic andesite lava flows. Elevation contours and heights of summits are shown in feet after the map of ONC G7 (Operational Navigation Charts G7 1987). In the hatched area elevations are greater than 17 000 feet.

Between 80°E and 84°E, several faults splay from the main branch of the Altyn Tagh fault (Peltzer, Tapponnier & Armijo 1989). The Altyn Tagh fault is composed of two faults near Puru. The northern fault lies along the contour of 3300 m at the northern foot of the Kunlun range, and the southern one lies along the contour of approximately 5700 m at its southern foot. The distance from the northern fault to Puru is 4 km and that from the southern one is 16 km.

The Puru area is in the Shasha geological province (Liushentage subprovince) (Xinjiang stratigraphic table compiling group 1981). This geological province is a unique area, where Cenozoic volcanics are exposed in the Tarim geological block. The basement was consolidated in the late Proterozoic. Overlying it are Precambrian through Pliocene platform clastics. Basic-to-intermediate volcanism occurred in the Pleistocene.

Flat-lying basaltic andesite lava flows are exposed north-east of Puru village. The sampling localities are adjacent to the road from 36°13′N, 81°31′E to 36°17′N, 81°28′E. They are located between 13 km and 18 km from the nearest strand of the Altyn Tagh fault. The lava sequence consists of two flows. The total thickness of the flows varies from 5 to 150 m. The lower one disappears in the southern part of the study area. The lower lava flow overlies a flat-lying layer of fluviatile gravels at site 15. The upper and lower lava flows intercalate with a volcaniclastic layer about 8 m in thickness at site 16. This layer shows an insignificant tilt of less than 2° with various strike azimuths at several points, so that the bedding attitude of both the lava flows is almost horizontal.

We have collected samples at three sites from the lower lava flow, and at 10 sites from the upper flow using a magnetic compass to measure the orientation of the samples. Each site typically comprised of seven independently oriented samples, distributed over a distance ranging up to 15 m. The present declination value at each sampling point was evaluated from a calculation using the International Geomagnetic Reference Field of 1989 (IAGA Division I Working Group 1 1987).

K-Ar DATING

K-Ar whole-rock dating was conducted for five samples following a petrographic study. Thin sections of samples were examined to determine their suitability for dating by this method, according to criteria discussed by Mankinen & Dalrymple (1972).

Samples for K-Ar analysis were prepared using the method described by Briggs *et al.* (1989). Groundmass-rich separates of 80–100 mesh size were used for the argon analysis, and a portion of each of the separates was further

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ground to use for potassium analysis. Analyses of potassium and argon, and calculation of ages and errors, were carried out according to the method described by Nagao *et al.* (1984) and Itaya *et al.* (1991). The decay constants of Steiger & Jäger (1977) were used in the age calculations.

The mean age and related uncertainty in determinations for each lava flow are shown in Table 1. The mean age is computed by assigning unit weight to each age datum. The uncertainty in age Δt at the 95 per cent confidence level is expressed as $\Delta t = t_c SN^{-1/2}$, where N is the number of age determinations from each lava flow, S is the variance of the ages, and t_c is the confidence coefficient at the 97.5 per cent point on the distribution of the Student t distribution with $N-1$ degrees of freedom. The age measurements for the upper lava flow ranges between 1.00 and 1.09 Ma. The mean age and 95 per cent confidence limit are 1.05 Ma and 0.11 Ma, respectively. The lower flow yields a mean age of 1.44 ± 0.06 Ma.

PALAEOMAGNETISM

In the laboratory, individual specimens, 25 mm in diameter and 25 mm long, were prepared from five samples from each site. Natural remanent magnetizations (NRMs) were measured with a spinner magnetometer. Intensity of the remanent magnetization for these samples ranges from 0.4 to 20 A m^{-1}.

All specimens were subjected to stepwise thermal demagnetization with 14 steps, up to 600 °C. Results for each specimen were plotted on orthogonal vector diagrams (Zijderveld 1967) to assess directional stability as well as on equal-area projections. Principal component analysis (Kirschvink 1980) was used to estimate the direction. In general, the rocks responded to thermal demagnetization in a straightward and uncomplicated way (Fig. 2). Except for a weak present-field overprint, none showed evidence of multiple magnetizations. Unblocking temperature ranges around 600 °C, indicating that this component is carried by titanium-poor magnetite.

Formation-mean directions for lavas

In all 13 sites characteristic directions for the single component could be isolated (Table 1). The mean direction of the characteristic directions for each site was calculated. No sites were discarded as unreliable, because the circle of 95 per cent confidence for each site is less than 8°.

Directions for both lower and upper lava flows are well

Table 1. Palaeomagnetic directions and K-Ar dating results of the basaltic andesites in the Puru area. Sampling sites are ordered from north to south. Site 13 and site 25 are estimated to be located at 36°17′N, 81°28′E and 36°13′N, 81°31′E, respectively. N is the number of field-oriented samples. The temperature range used to calculate characteristic direction by principal component analysis is shown in °C. D and I are declination and inclination, respectively: k is the Fisher precision parameter; α_{95} is the radius of the cone of 95 per cent confidence about a mean direction. Seven mean directions are calculated for the upper lava flow on the basis of data from four to 10 sampling sites between 13 km and 18 km from the fault. The range of sampling sites used to calculate the mean direction is shown in km. Unit weight is assigned to each lava flow when an average palaeomagnetic direction for the basaltic andesite of the Puru area is calculated. VGP is the virtual geomagnetic pole. Ages are calculated from the following decay and abundance constants: $\lambda_\varepsilon = 0.581 \times 10^{-10}\,yr^{-1}$; $\lambda_\beta = 4.962 \times 10^{-10}\,yr^{-1}$; $^{40}K/K = 1.167 \times 10^{-4}$ mol mol^{-1} (Steiger & Jäger 1977). The Altyn Tagh fault is assumed to pass through a point at 36°07′N, 81°30′E with a strike of N80°E in the Puru area.

Site	N	Temperature range (°C)	D(°)	I(°)	k	α_{95}(°)	K-Ar (Ma)	Potassium (wt %)	$^{40}Ar_{rad}$ (x 10^{-8} cm^3/g)	Ar_{air}/Ar_{total} (%)	Distance from the Altyn Tagh fault (km)
[Upper lava flow]											
13	6	0-600	-170.4	-50.3	737.6	2.5					18
17	6	0-600	-164.3	-52.2	338.0	3.6					17.4
18	7	200-600	-165.1	-51.6	300.4	3.5	1.05±0.05	3.299±0.066	13.5±0.5	66.3	17.1
19	7	200-600	-160.0	-50.0	169.8	4.6	1.00±0.03	3.547±0.071	13.8±0.3	40.1	16.7
20	6	100-600	-169.8	-45.0	239.3	4.3					16.4
21	6	100-600	-170.3	-53.3	90.7	7.1	1.09±0.04	3.456±0.069	14.6±0.4	51.5	16.1
23	6	0-600	-163.4	-52.8	404.2	3.3					16.0
24	6	100-600	-168.1	-49.8	335.7	3.7					15.8
27	6	100-600	-164.8	-51.9	526.1	2.9					14
25	5	100-600	-173.7	-52.3	1004.1	2.4					13
mean	10	13km-18km	-167.0	-51.0	525.1	2.1	1.05±0.11				
	9	14km-18km	-166.3	-50.8	574.1	2.2					
	8	15.8km-18km	-166.5	-50.7	515.3	2.4					
	7	16km-18km	-166.2	-50.8	452.4	2.8					
	6	16.1km-18km	-166.7	-50.5	416.4	3.3					
	5	16.4km-18km	-166.0	-49.9	415.3	3.8					
	4	16.7km-18km	-164.9	-51.1	779.5	3.3					
[Lower lava flow]											
15	5	400-600	170.0	-57.3	1087.2	2.3	1.44±0.04	3.313±0.066	18.5±0.4	46.4	17.6
16	7	100-600	174.4	-55.3	393.5	3.0					17.5
22	6	100-600	171.5	-54.3	6068.2	0.9	1.43±0.04	3.293±0.066	18.2±0.4	49.0	16.2
mean	3		172.0	-55.6	1679.4	3.0	1.44±0.06				

[Upper and Lower lava flows]				VGP (North) Lat.	Long.
MEAN	2	-176.9	-53.8	86.8°N	153.1°W

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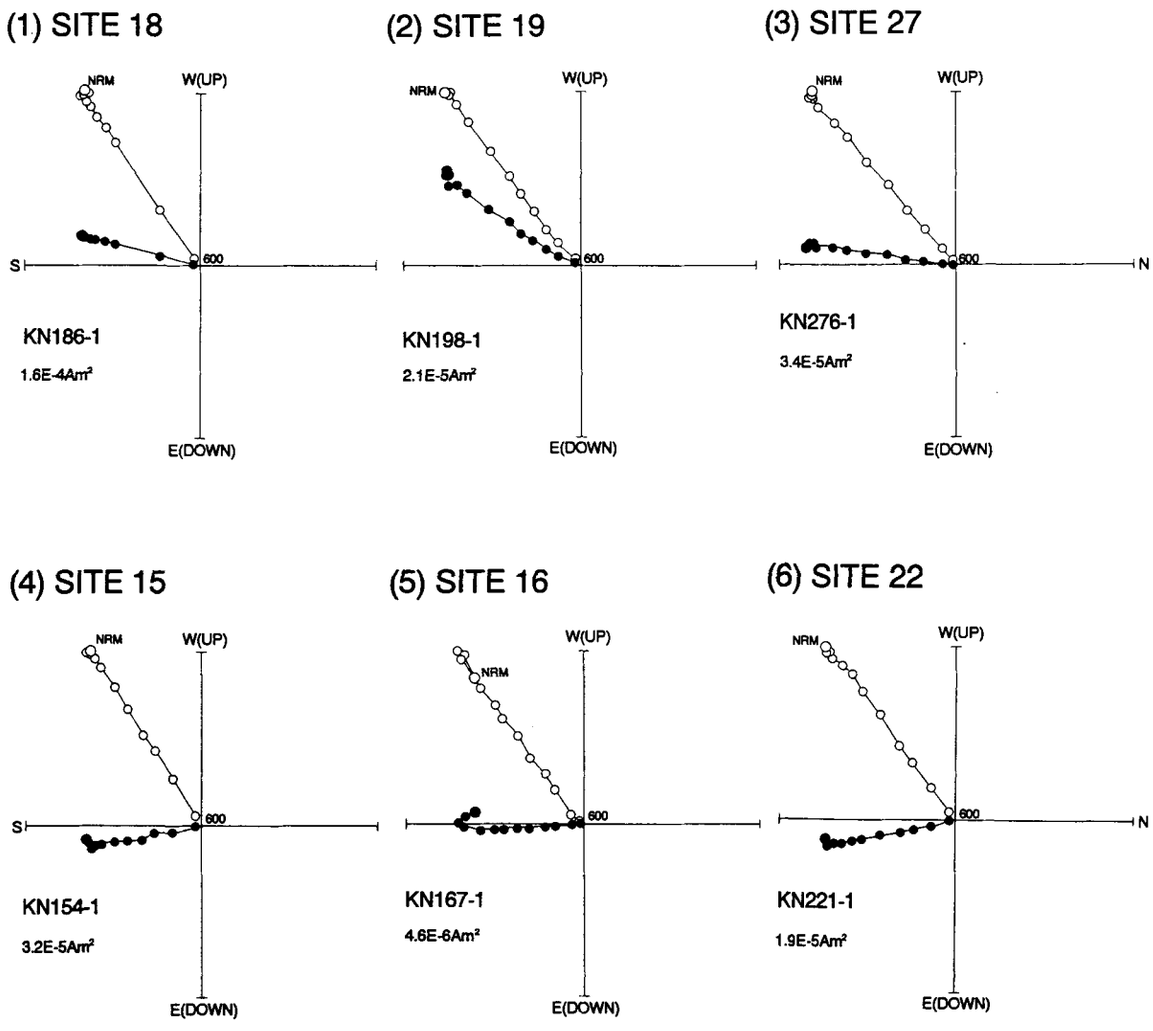


Figure 2. Orthogonal projections of the magnetization vector endpoints during thermal demagnetization experiments for specimens from the upper (1–3) and lower lava flows (4–6). Open symbols show the magnetic vector projected onto the vertical plane, solid symbols onto the horizontal plane. Numbers by the symbols are temperature (°C).

grouped and all of reversed magnetic polarity (Fig. 3). The palaeomagnetic mean direction for the lower lava flow is $D = 172.0°$, $I = -55.6°$, $\alpha_{95} = 3.0°$, while that for the upper one is $D = -167.0°$, $I = -51.0°$, $\alpha_{95} = 2.1°$.

Declination versus distance from the fault

Declinations as a function of distance from the fault are shown in Fig. 4. For the upper lava flow, insignificant variation in declination is observed for nine sites between 14 km and 18 km from the fault, whereas a site at 13 km shows a significant difference. The best-fitting line passing through the nine sites (dotted line) has a negative and small gradient of $-0.41° \text{ km}^{-1}$. This indicates that the declination value does not change between 14 km and 18 km. Absence of the internal variation in declination is supported through calculation of the palaeomagnetic mean directions. The mean direction of all nine sites from 14 km to 18 km is quite similar to those calculated on the basis of 4, 5, 6, 7 or 8 data from 16.1 km to 18 km (see Table 1).

A significant eastward shift in declination is observed for the sampling site 13 km from the fault. The palaeomagnetic direction for this site is significantly different at the 95 per cent level from the mean direction for the remaining nine sites between 14 km and 18 km (Fig. 3c). Amount of the deflection at 13 km is evaluated to be $7.4° \pm 5.2°$ counter-clockwise, where the uncertainty in declination is given by $\sin^{-1}(\sin \alpha_{95}/\cos I)$. The best-fitting line analysis also indicates that the declination at 13 km is distinct from the others (Fig. 4). The observed declination at 13 km ($D_{ob} = -173.7°$) reveals significant counter-clockwise deflection of 8.8° from the expected direction ($D_{ex} = -164.9° \pm 3.8°$ (1σ)) which is obtained by extrapolation of the best-fitting line to this site. If the datum of site 25 is incorporated into those of the remaining nine sites to obtain another best-fitting line (solid line in Fig. 4), the sign of gradient of the line changes from a negative to a positive and fairly large gradient of $0.88° \text{ km}^{-1}$ appears. This implies that the datum at 13 km is heterogeneous from that of the others.

There is no significant variation in declination for the lower lava flow.

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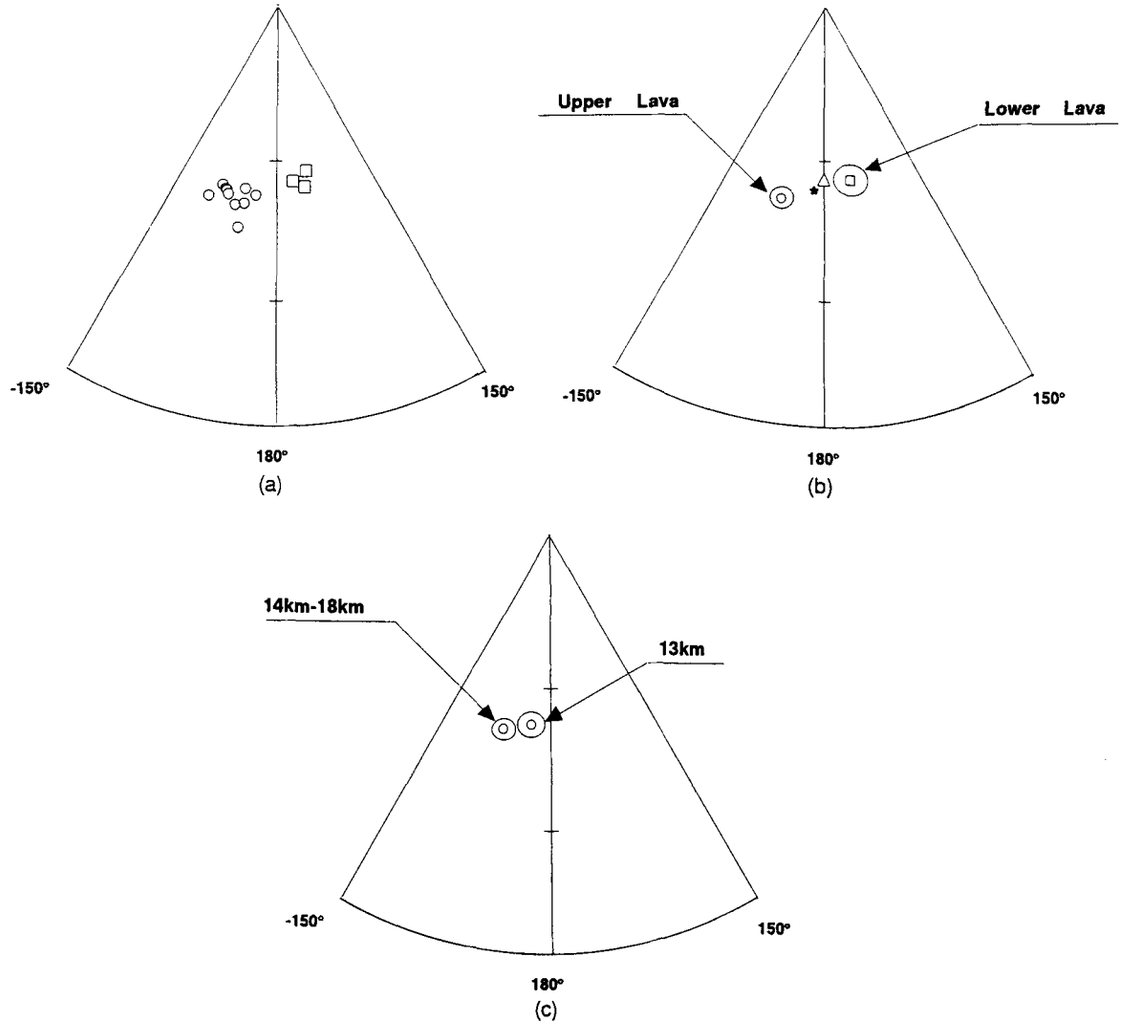


Figure 3. Equal-area projections of the site mean directions for the Pleistocene lava flows. (a) Mean direction of each site for the upper (lower) lava flow is shown by a circle (square). (b) Mean directions with 95 per cent confidence circles for the upper and lower lava flows are drawn as circles and squares, respectively. Triangle: present axial dipole field direction. Asterisk: present geomagnetic field direction calculated from IGRF 1989 (IAGA Division I Working Group 1 1987). (c) Mean directions with 95 per cent confidence circle for the upper lava flow: a mean direction based on nine data points from the sampling sites between 14 km and 18 km from the Altyn Tagh fault, and the mean direction of the sampling site at 13 km from the fault (site 25). All symbols are plotted on an upper hemisphere projection.

DISCUSSION

Small dispersion in the NRM directions ensures a high reliability for our palaeomagnetic data: the α_{95} of the mean direction for the upper lava is only 2.1°. Orientation of the samples was by magnetic compass, but the values do not appear to have been affected by the magnetism of the rocks, despite the variable NRM intensity. The small dispersion supports the inference that in the Puru area this basaltic andesite lava flowed smoothly, without rotation about a vertical axis after it cooled below blocking temperature (Sakamoto 1992).

The lava flows of the Puru area record the palaeomagnetic direction of the Matuyama chron. Reversed polarity observed in lava flows with ages of 1.05 Ma and 1.44 Ma is consistent with the geomagnetic polarity time-scale of Harland *et al.* (1990).

Palaeomagnetic directions for the lower and upper lavas are statistically distinguished: the difference in palaeomagnetic directions between the data sets of these lava flows is 13.3°, with an uncertainty (95 per cent confidence) of 3.7°. This difference probably reflects the secular variation of the ancient geomagnetic field in the Matuyama chron. The probability is quite low for either lava to have acquired their magnetizations during a polarity transition, because no polarity reversals occurred in the periods between 0.97 Ma (the lower Jaramillo polarity transition) and 1.62 Ma (the upper Gilsa polarity transition) except for the polarity reversals of the Cobb mountain subchron at about 1.1 Ma (Champion, Lanphere & Kuntz 1988; Hyodo, Sunata & Susanto 1992).

It is hard to detect a small-scale block rotation of only 10° from the mean directions for the lava flows of the Puru area. A clockwise rotation of $8.0° \pm 5.3°$ may be suggested according to the mean direction for the lower lava flow, assuming 180° as an expected declination value. A

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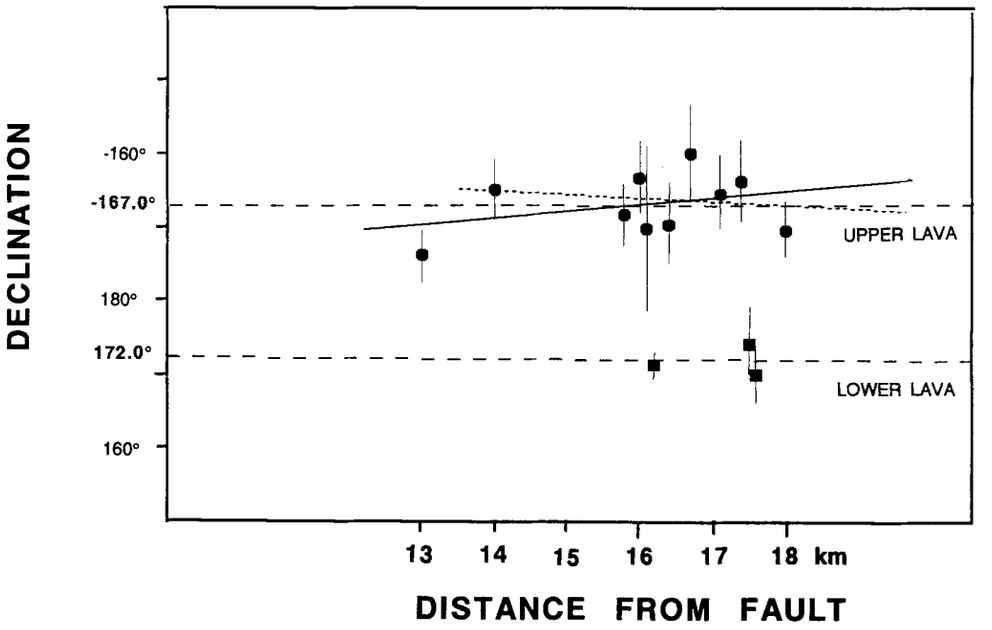


Figure 4. Declination versus distance of the sampling site from the Altyn Tagh fault in the Puru area. Data for the upper (lower) lava flow are shown by circles (squares). Mean declination values for the upper and lower lava flows (−167.0° and 172.0°) are shown by dashed lines. Two best-fitting lines (a unit-weighted least-squares fit) are determined for the upper lava flow using all 10 data points between 13 km and 18 km from the fault (solid line), and using the nine data points between 14 km and 18 km (dotted line). The former reveals a positive gradient of 0.88° km^{-1}, whereas the latter has a negative and small gradient of −0.41° km^{-1}.

counter-clockwise rotation of 13° can be, in contrast, implied from the upper lava flow. These complicated estimates in rotation are only ascribed to the inadequate sampling of the palaeomagnetic directions for averaging secular variation of the geomagnetic field. We claim, however, that the Puru area has not experienced large rotational motion of more than 60°, as has been reported from the areas along the San Andreas fault in southern California and the Alpine fault in New Zealand (Luyendyk *et al.* 1985; Walcott 1984; Lamb 1988).

Declination change as a function of distance from a fault provides an alternative method for detecting a small-scale rotation. The present study suggests that a counter-clockwise palaeomagnetic rotation of 7.4 ± 5.2° occurred in the area 13 km from the fault with respect to the area between 14 km and 18 km from the fault. This difference in declination is not ascribed to secular variation of the geomagnetic field. Little record of secular variation is preserved in the NRM directions of the upper lava flow, because the thickness of this flow is too thin (less than 150 m) to record a large change in direction due to secular variation longer than 200 years (e.g. Turcotte & Schubert 1982). The palaeomagnetic rotation is, rather, ascribable to tectonic rotation due to left-lateral slip of the Altyn Tagh fault (Fig. 5). Several models can explain the counter-clockwise rotation along left-lateral slip faults (e.g. Nelson & Jones 1987): (a) the deformation of a continuum, (b) a small block model with variable internal rotation, and (c) a large block-rotation model with internal antithetic shear (book-shelf kinematics). The area 13 km from the fault has probably experienced tectonic rotation with respect to the interior part of the Tarim craton.

Our palaeomagnetic study does not detect a large rotation of more than 60° outside the area 13 km from the fault. The large deformation is probably limited to a narrow zone less than 13 km from the fault. This aspect of the rotation was not predicted. Since the slip rate of the Altyn Tagh fault is 30 mm per year (Peltzer *et al.* 1989), fairly large rotation (~60°) was expected at a distance of 13 km with more than 30 km displacement during the last 1 myr (Fig. 5). The narrow deformation zone along the Altyn Tagh fault contrasts with that of the San Andreas fault and the Alpine fault, where the deformed zones are spread out over a width of about 100–200 km (Howard *et al.* 1978; Crowell 1979; Thatcher 1979; Lamb 1988; Luyendyk *et al.* 1985; Jackson & Molnar 1990).

The narrow region of disturbance along the Altyn Tagh fault is quite similar to the features of transform faults on the Mid-Atlantic Ridge and the East Pacific Rise in the ocean (Gallo, Fox & Macdonald 1986; Pockalny, Detrick & Fox 1988; Carbotte & Macdonald 1992). Fabrics parallel to the ridge axis (ridge-flank topography and Vine Mathews magnetic-anomaly promontories) beyond 6 km from the fault have not been deformed by strike-slip movements for about 2 myr for either fast (110 mm yr^{-1}) or slow (30 mm yr^{-1}) slipping at the Clipperton Transform Fault (Gallo, Fox & Macdonald 1986).

The narrow deformation zone along the Altyn Tagh fault is attributed to the nature of the continental lithosphere of the Tarim Craton and/or the nature of the strike-slip movement of the Altyn Tagh fault. The continental crust near the fault exhibits a non-rigid behaviour (either as a non-Newtonian fluid with low apparent viscosity Fig. 5a or as a brittle material Fig. 5b), similar to the oceanic lithosphere at a plate margin (Gallo *et al.* 1986; Carbotte & Macdonald 1992). An alternative explanation is the smooth

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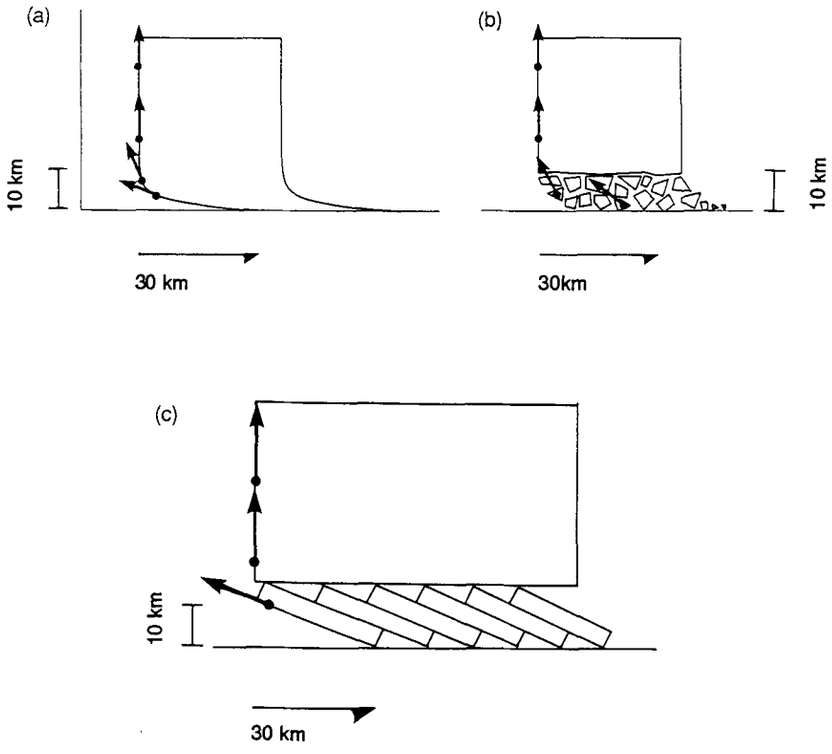


Figure 5. Possible mechanisms of deformation along the Altyn Tagh fault. Counter-clockwise rotation can take place in the strip of crust along the fault in the following three models (after Nelson & Jones 1987), when the Altyn Tagh fault is displaced in the sinistral sense by 30 km during the last 1 myr Arrows represent the directions of palaeomagnetic vectors. (a) Rotation of a deforming continuum. The large displacement and large rotation are restricted to within 10 km of the fault, as the crustal strip behaves as a non-Newtonian fluid with low apparent viscosity. (b) Small block model with variable internal rotation. The crust behaves as a brittle material. The crustal strip within 13 km of the fault is broken into pieces which rotate independently. (c) Large block-rotation model (bookshelf kinematics). The amount of rotation inferred from palaeomagnetic data is the same everywhere in the rotating block.

slip of the Altyn Tagh fault. Low stress is inferred for the Altyn Tagh fault during the last 1 myr, because large earthquakes are rarely observed along the fault (e.g. China Distribution of Seismicity 1990; Xu *et al.* 1992).

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